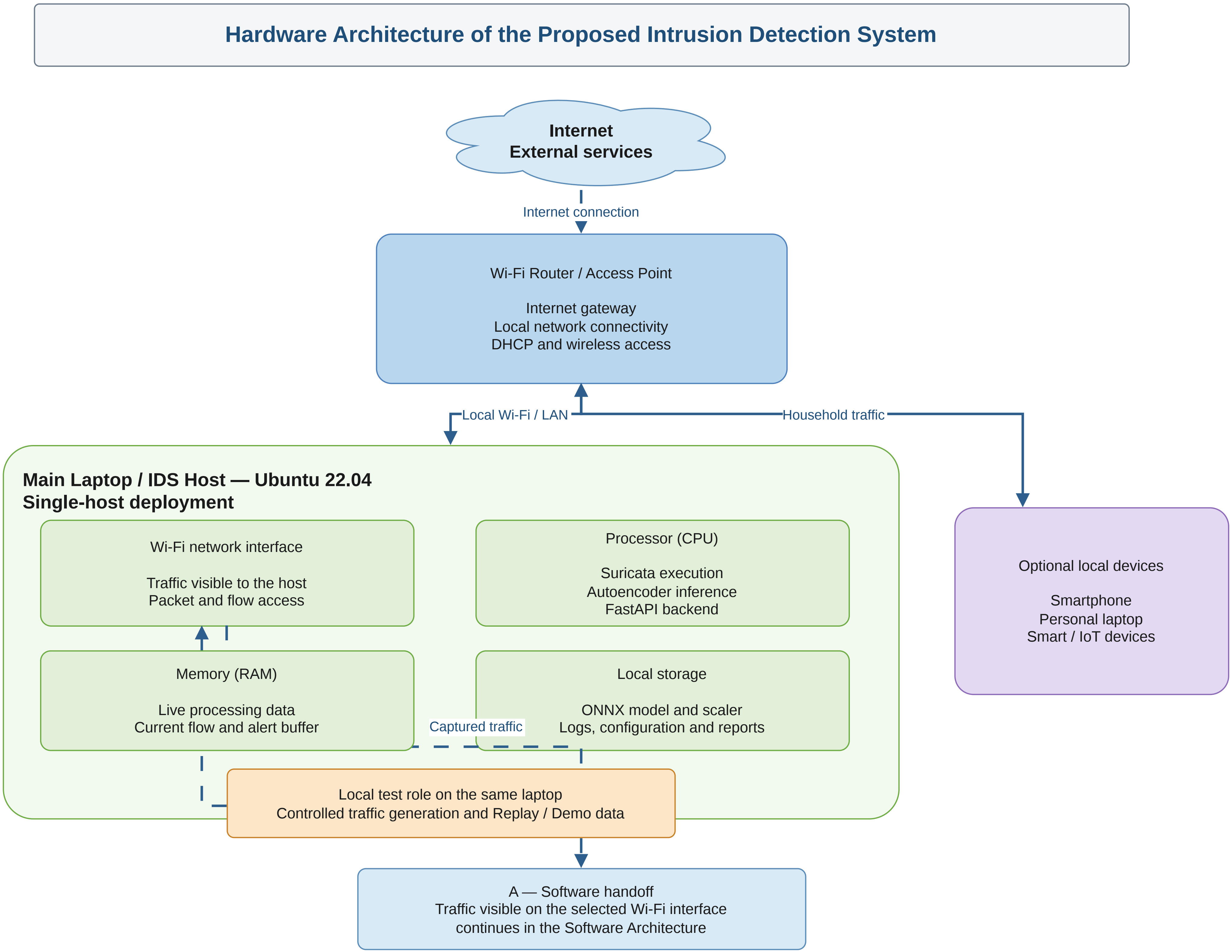
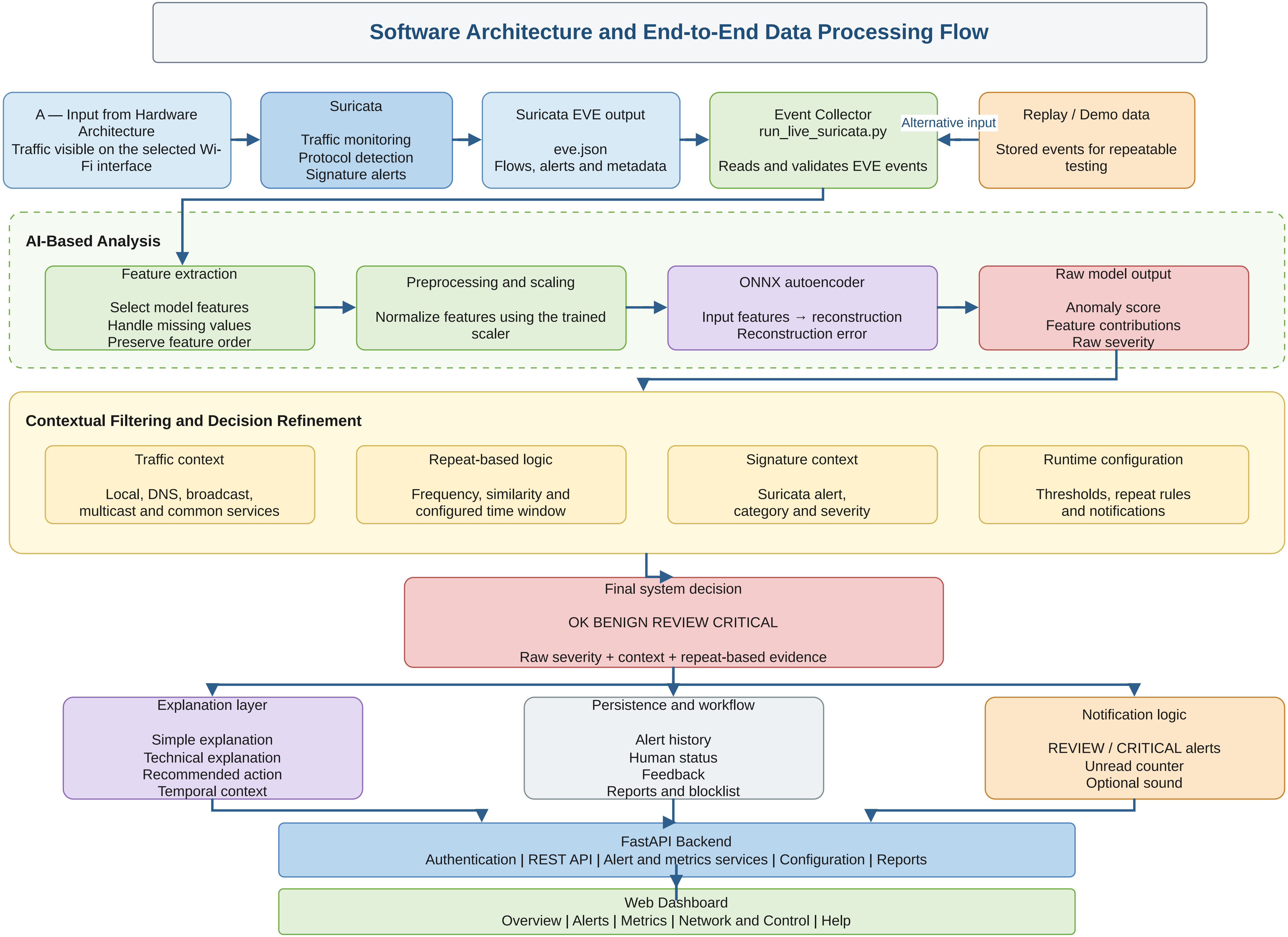
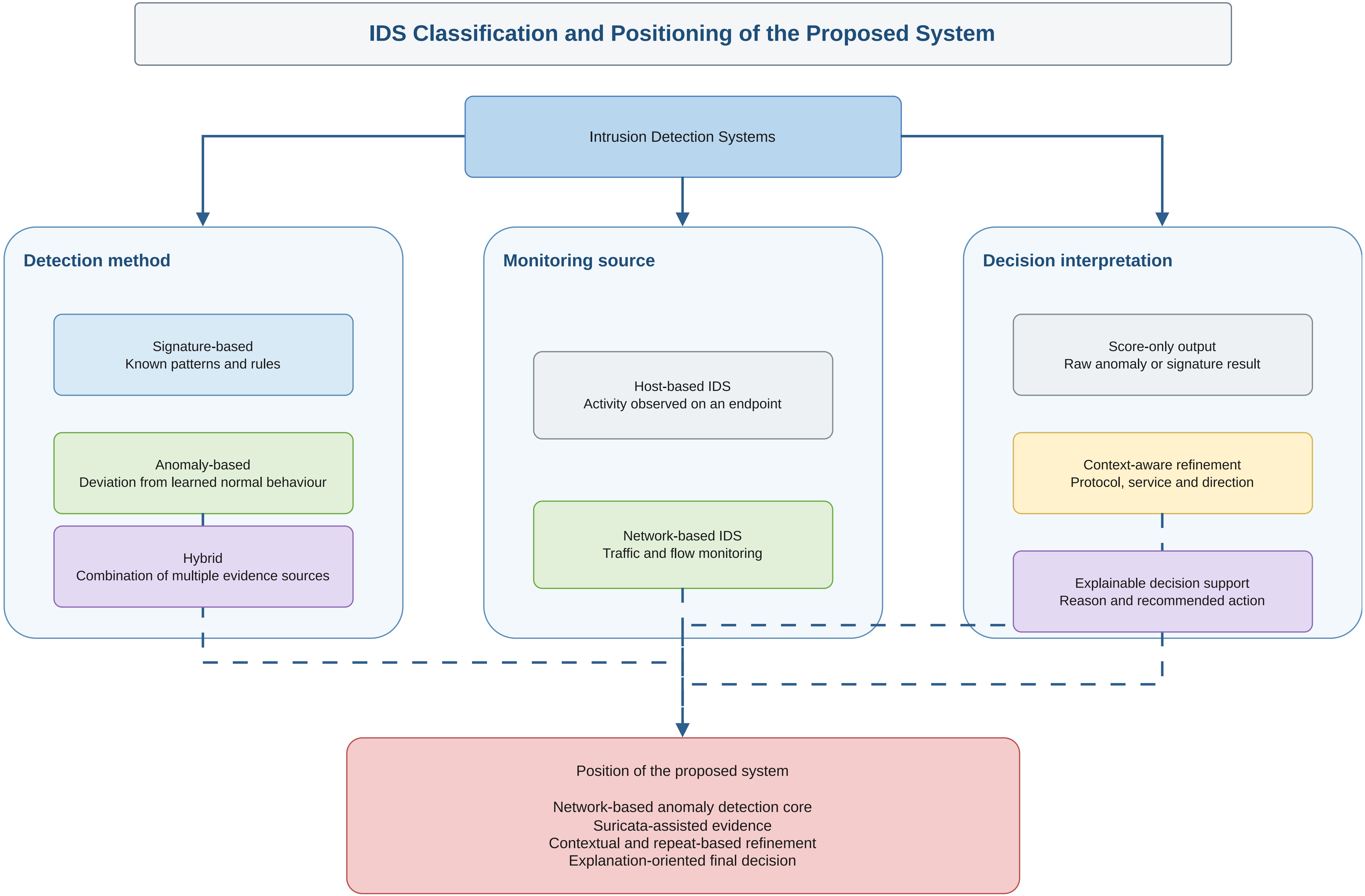
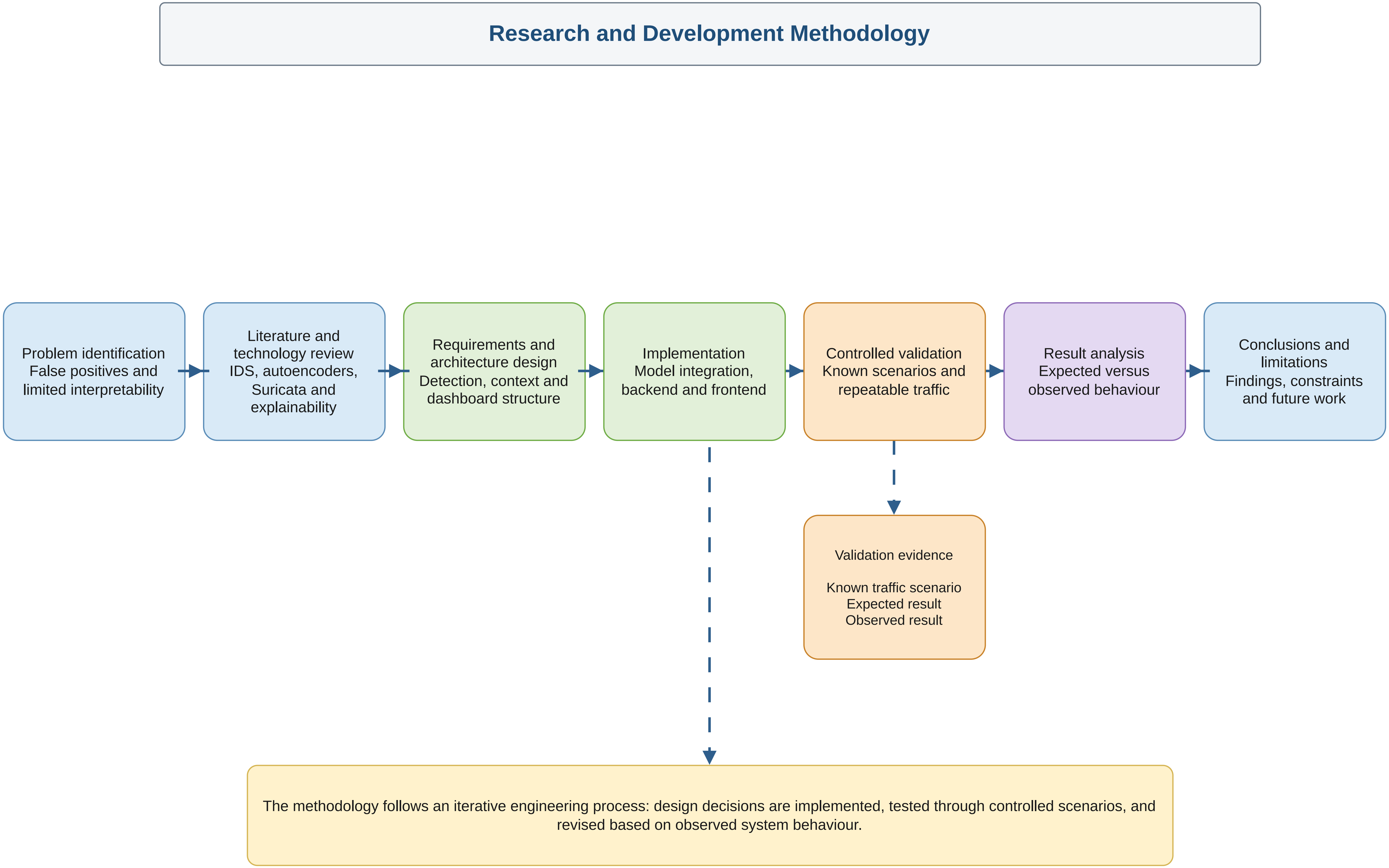
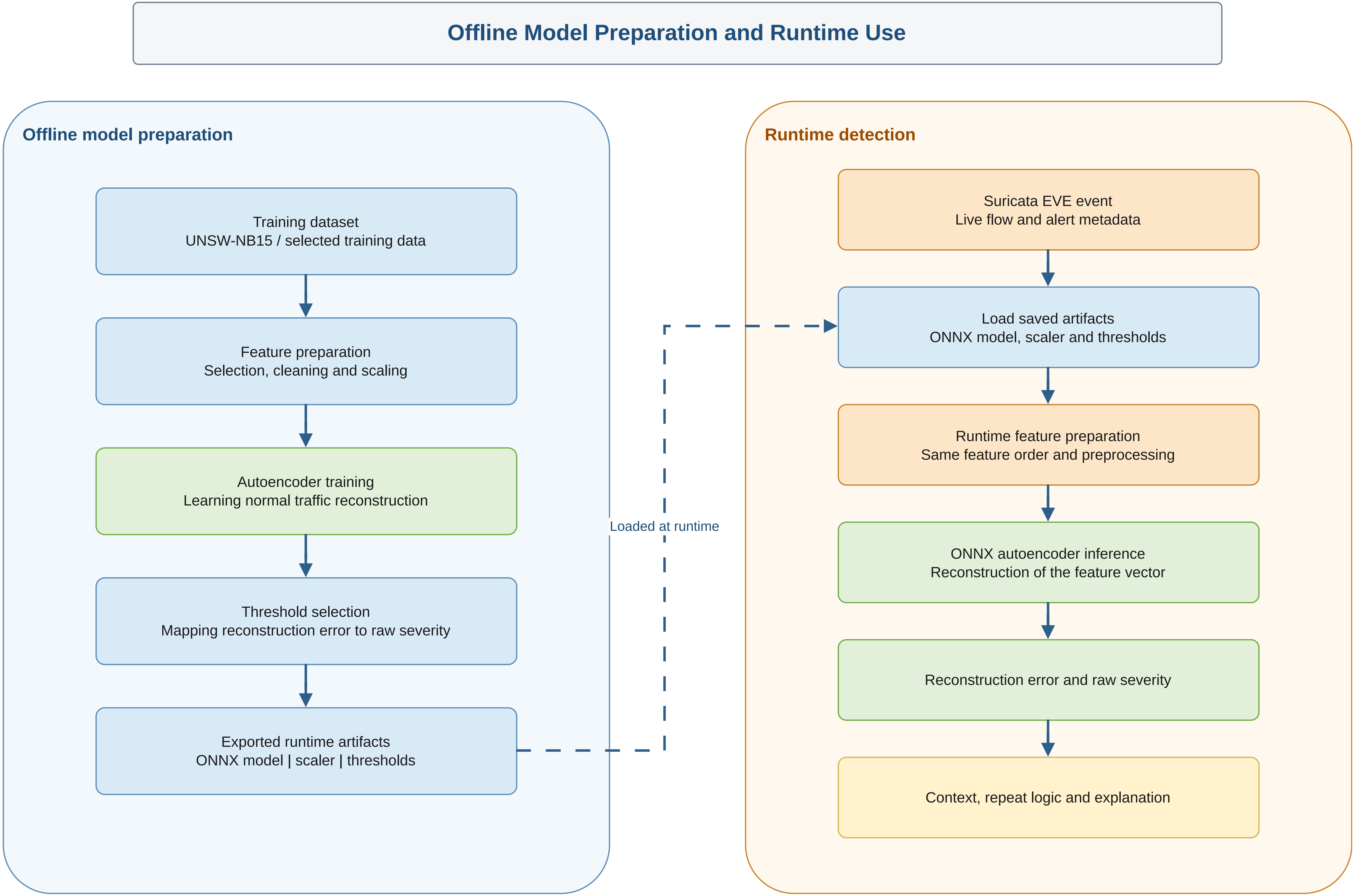
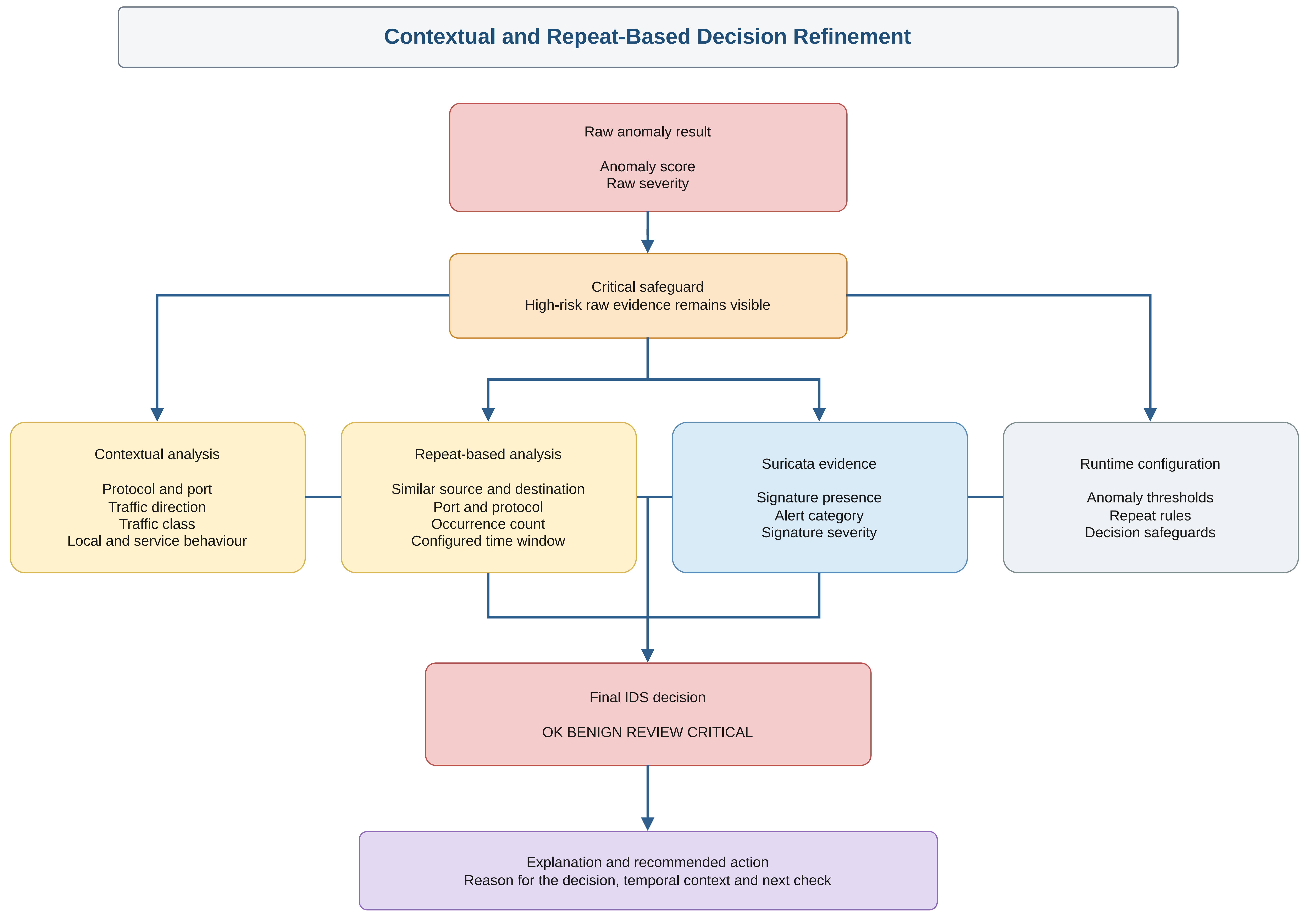

Hardware Architecture of the Proposed Intrusion Detection System
Internet
External services
Internet connection
Wi-Fi Router / Access Point
Internet gateway
Local network connectivity
DHCP and wireless access
Local Wi-Fi / LAN
Household traffic
Main Laptop / IDS Host — Ubuntu 22.04
Single-host deployment
Wi-Fi network interface
Traffic visible to the host
Packet and flow access
Processor (CPU)
Suricata execution
Autoencoder inference
FastAPI backend
Memory (RAM)
Live processing data
Current flow and alert buffer
Local storage
ONNX model and scaler
Logs, configuration and reports
Captured traffic
Local test role on the same laptop
Controlled traffic generation and Replay / Demo data
Optional local devices
Smartphone
Personal laptop
Smart / IoT devices
A — Software handoff
Traffic visible on the selected Wi-Fi interface
continues in the Software Architecture


Software Architecture and End-to-End Data Processing Flow
A — Input from Hardware Architecture
Traffic visible on the selected Wi-Fi interface
Suricata
Traffic monitoring
Protocol detection
Signature alerts
Suricata EVE output
eve.json
Flows, alerts and metadata
Event Collector
run_live_suricata.py
Reads and validates EVE events
Alternative input
Replay / Demo data
Stored events for repeatable testing
AI-Based Analysis
Feature extraction
Select model features
Handle missing values
Preserve feature order
Preprocessing and scaling
Normalize features using the trained scaler
ONNX autoencoder
Input features → reconstruction
Reconstruction error
Raw model output
Anomaly score
Feature contributions
Raw severity
Contextual Filtering and Decision Refinement
Traffic context
Local, DNS, broadcast, multicast and common services
Repeat-based logic
Frequency, similarity and configured time window
Signature context
Suricata alert, category and severity
Runtime configuration
Thresholds, repeat rules and notifications
Final system decision
OK BENIGN REVIEW CRITICAL
Raw severity + context + repeat-based evidence
Explanation layer
Simple explanation
Technical explanation
Recommended action
Temporal context
Persistence and workflow
Alert history
Human status
Feedback
Reports and blocklist
Notification logic
REVIEW / CRITICAL alerts
Unread counter
Optional sound
FastAPI Backend
Authentication | REST API | Alert and metrics services | Configuration | Reports
Web Dashboard
Overview | Alerts | Metrics | Network and Control | Help

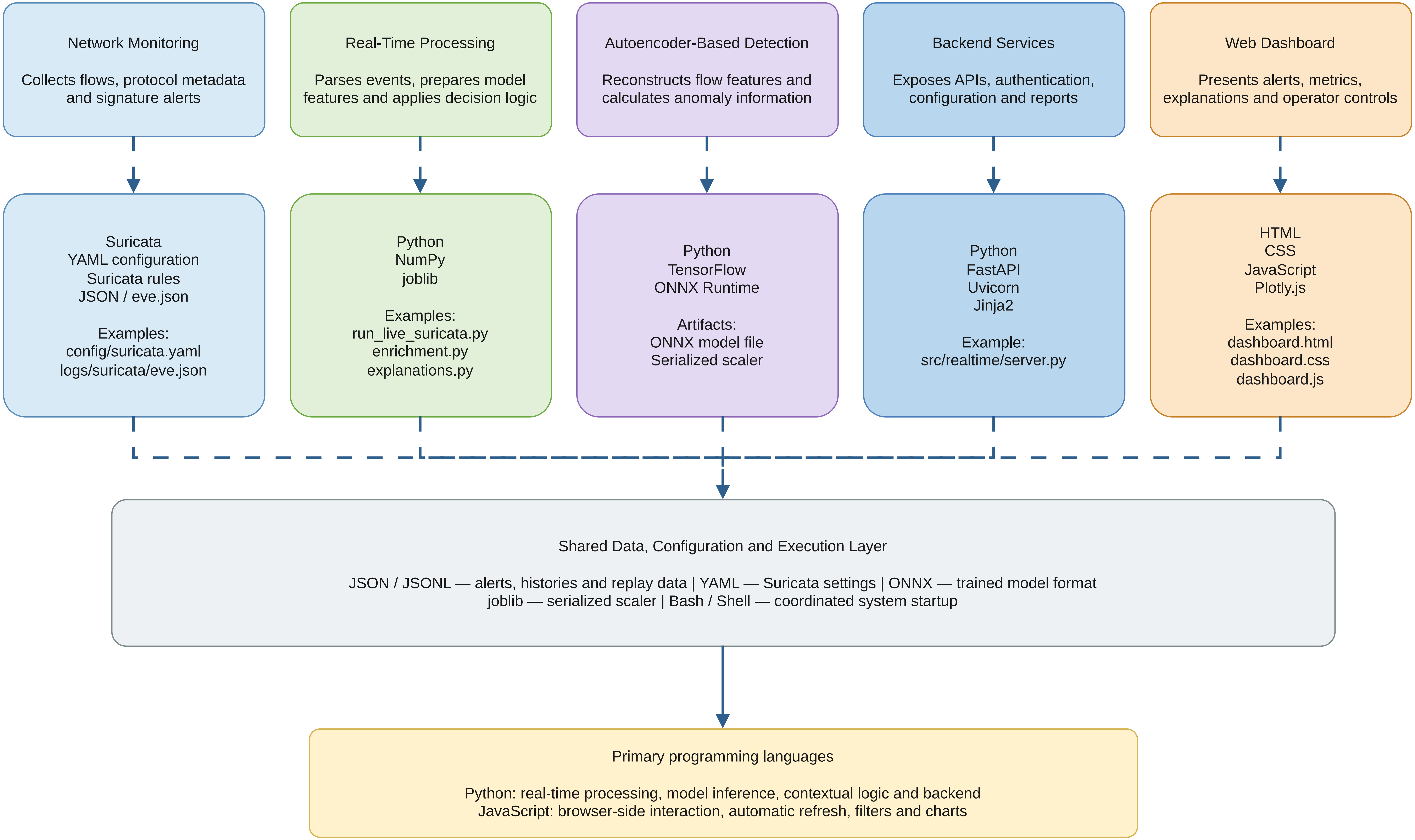

IDS Classification and Positioning of the Proposed System
Intrusion Detection Systems
Detection method
Signature-based
Known patterns and rules
Anomaly-based
Deviation from learned normal behaviour
Hybrid
Combination of multiple evidence sources
Monitoring source
Host-based IDS
Activity observed on an endpoint
Network-based IDS
Traffic and flow monitoring
Decision interpretation
Score-only output
Raw anomaly or signature result
Context-aware refinement
Protocol, service and direction
Explainable decision support
Reason and recommended action
Position of the proposed system
Network-based anomaly detection core
Suricata-assisted evidence
Contextual and repeat-based refinement
Explanation-oriented final decision

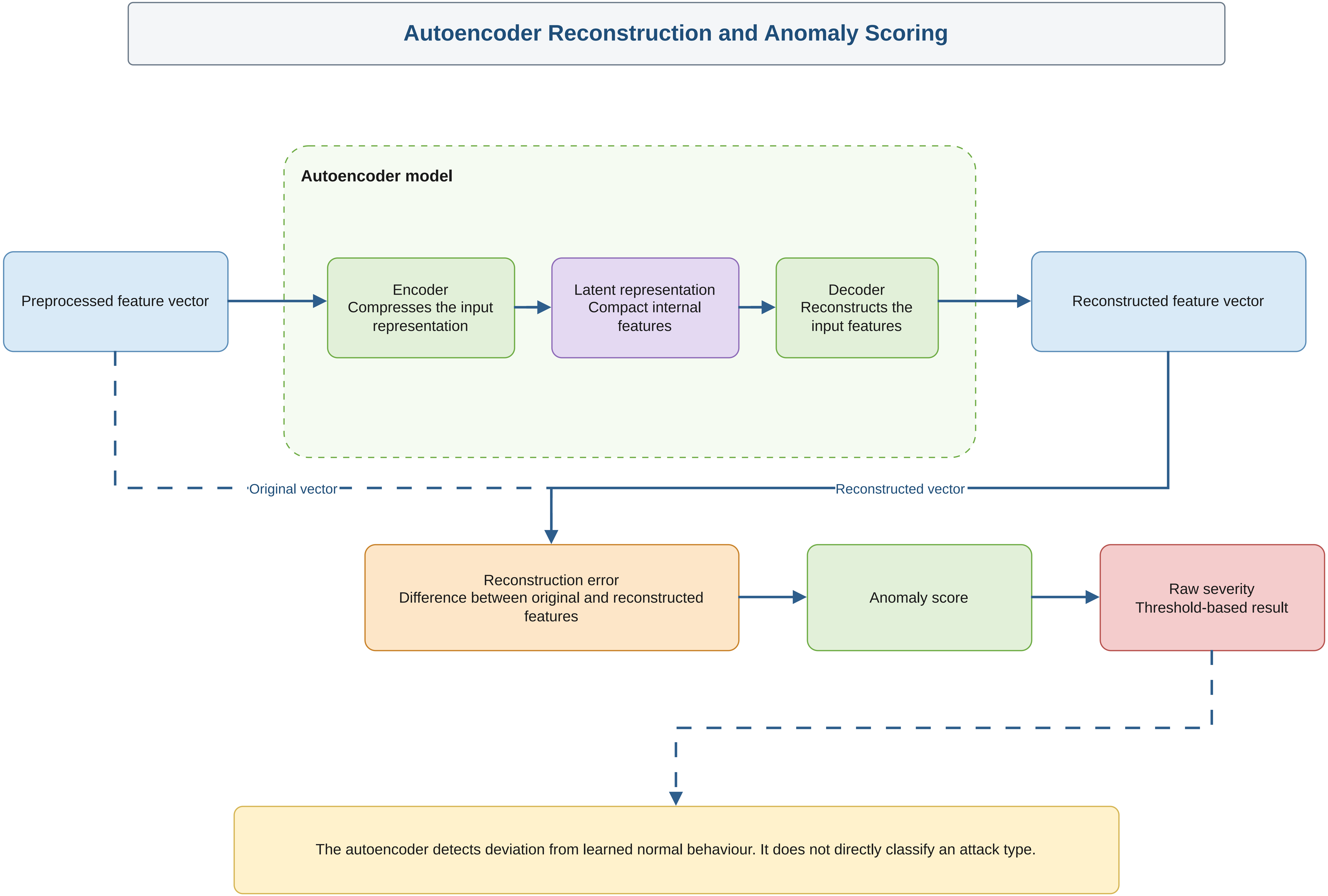

Research and Development Methodology
Problem identification
False positives and limited interpretability
Literature and technology review
IDS, autoencoders, Suricata and explainability
Requirements and architecture design
Detection, context and dashboard structure
Implementation
Model integration, backend and frontend
Controlled validation
Known scenarios and repeatable traffic
Result analysis
Expected versus observed behaviour
Conclusions and limitations
Findings, constraints and future work
Validation evidence
Known traffic scenario
Expected result
Observed result
The methodology follows an iterative engineering process: design decisions are implemented, tested through controlled scenarios, and revised based on observed system behaviour.


Offline Model Preparation and Runtime Use
Offline model preparation
Training dataset
UNSW-NB15 / selected training data
Feature preparation
Selection, cleaning and scaling
Autoencoder training
Learning normal traffic reconstruction
Threshold selection
Mapping reconstruction error to raw severity
Exported runtime artifacts
ONNX model | scaler | thresholds
Loaded at runtime
Runtime detection
Suricata EVE event
Live flow and alert metadata
Load saved artifacts
ONNX model, scaler and thresholds
Runtime feature preparation
Same feature order and preprocessing
ONNX autoencoder inference
Reconstruction of the feature vector
Reconstruction error and raw severity
Context, repeat logic and explanation

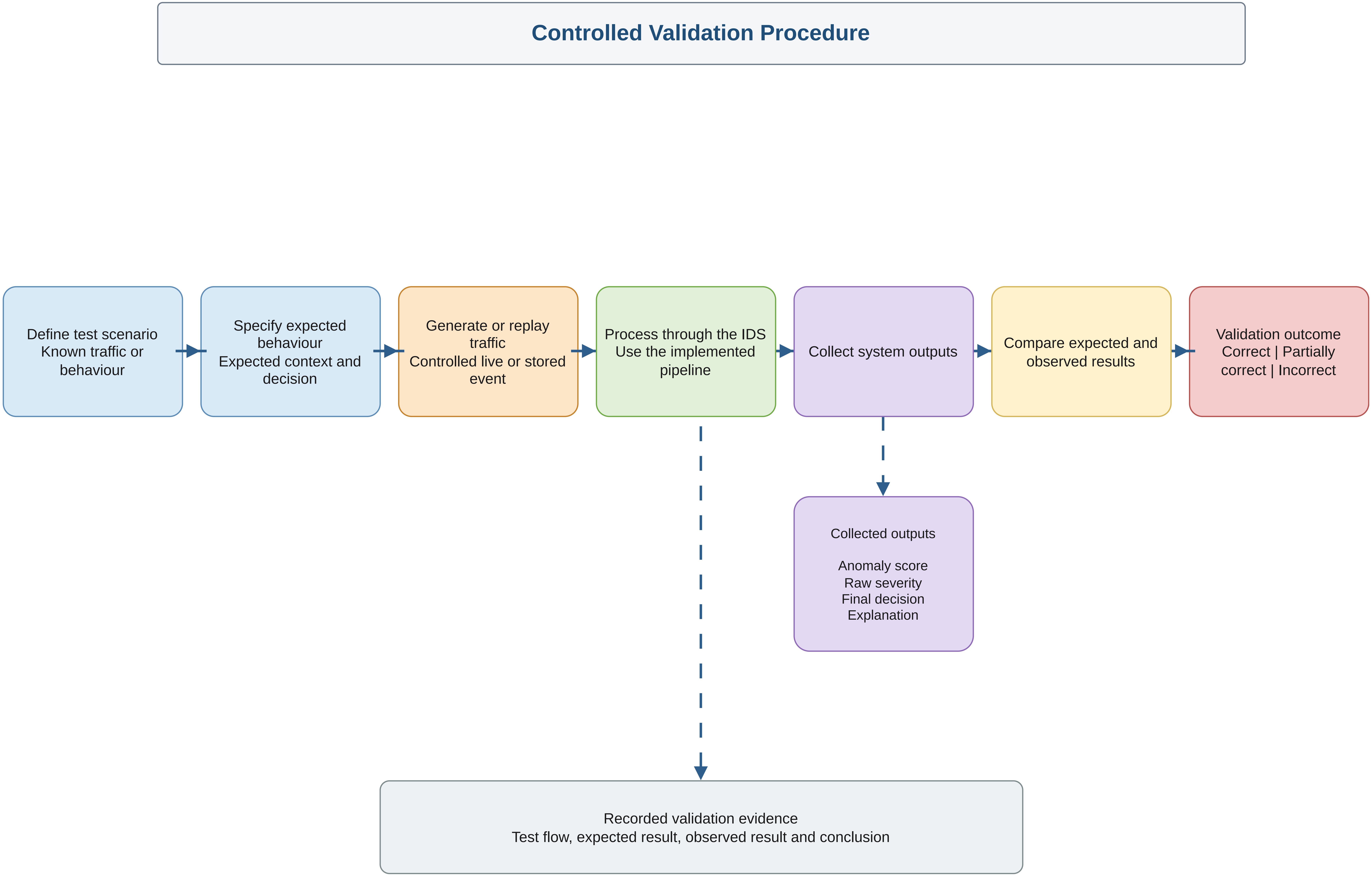

Contextual and Repeat-Based Decision Refinement
Raw anomaly result
Anomaly score
Raw severity
Critical safeguard
High-risk raw evidence remains visible
Contextual analysis
Protocol and port
Traffic direction
Traffic class
Local and service behaviour
Repeat-based analysis
Similar source and destination
Port and protocol
Occurrence count
Configured time window
Suricata evidence
Signature presence
Alert category
Signature severity
Runtime configuration
Anomaly thresholds
Repeat rules
Decision safeguards
Final IDS decision
OK BENIGN REVIEW CRITICAL
Explanation and recommended action
Reason for the decision, temporal context and next check

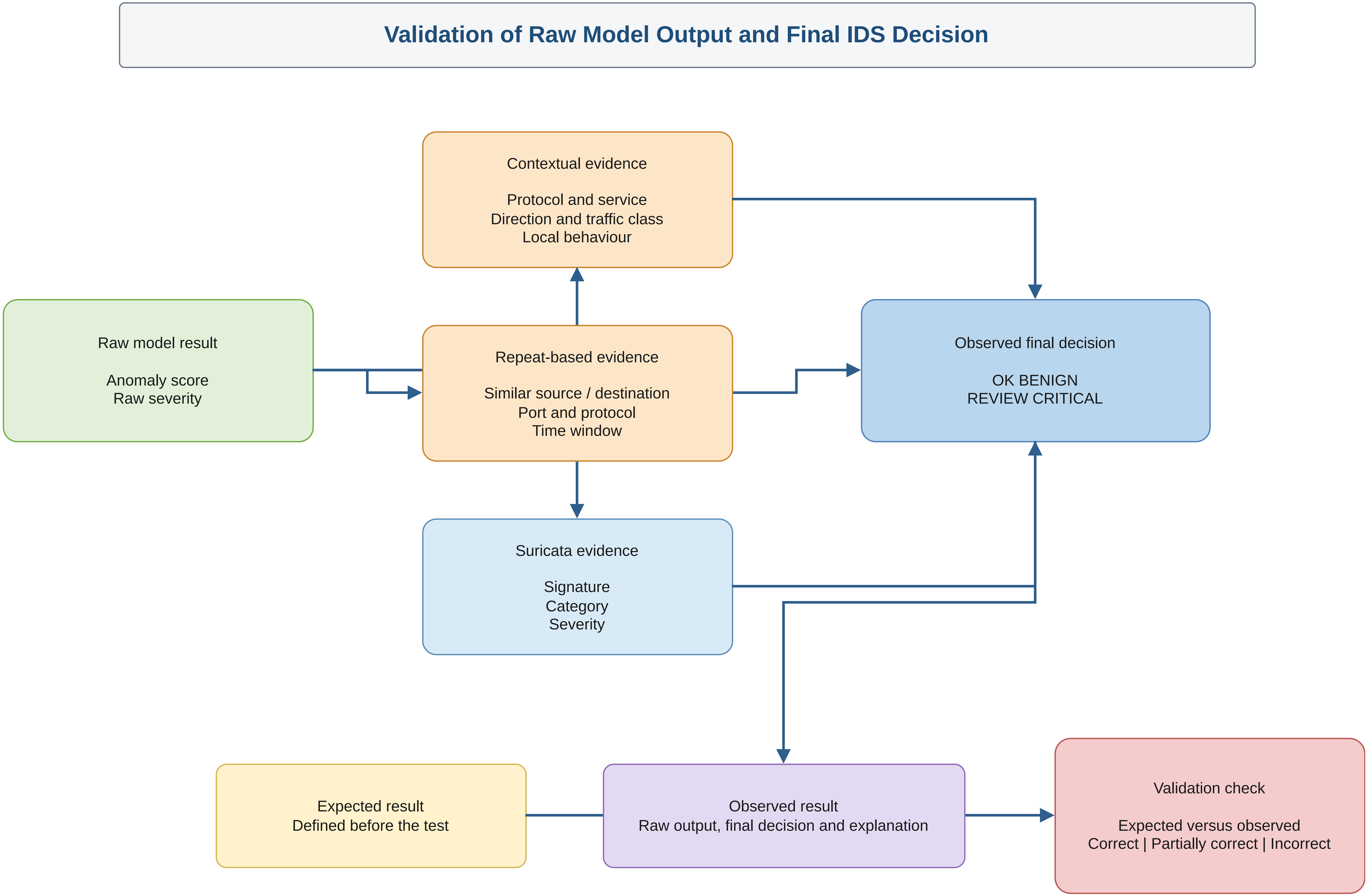
Technologies, Languages and Implementation Components





Autoencoder Reconstruction and Anomaly Scoring
Autoencoder model
Preprocessed feature vector
Encoder
Compresses the input representation
Latent representation
Compact internal features
Decoder
Reconstructs the input features
Reconstructed feature vector
Original vector
Reconstructed vector
Reconstruction error
Difference between original and reconstructed features
Anomaly score
Raw severity
Threshold-based result
The autoencoder detects deviation from learned normal behaviour. It does not directly classify an attack type.


Controlled Validation Procedure
Define test scenario
Known traffic or behaviour
Specify expected behaviour
Expected context and decision
Generate or replay traffic
Controlled live or stored event
Process through the IDS
Use the implemented pipeline
Collect system outputs
Compare expected and observed results
Validation outcome
Correct | Partially correct | Incorrect
Collected outputs
Anomaly score
Raw severity
Final decision
Explanation
Recorded validation evidence
Test flow, expected result, observed result and conclusion


Validation of Raw Model Output and Final IDS Decision
Contextual evidence
Protocol and service
Direction and traffic class
Local behaviour
Raw model result
Anomaly score
Raw severity
Repeat-based evidence
Similar source / destination
Port and protocol
Time window
Observed final decision
OK BENIGN
REVIEW CRITICAL
Suricata evidence
Signature
Category
Severity
Expected result
Defined before the test
Observed result
Raw output, final decision and explanation
Validation check
Expected versus observed
Correct | Partially correct | Incorrect